Case Systems and Grammatical Relations

Case Systems, Grammatical Relations, and Alignment Typology

Historical Foundations: From Pāṇini to Jakobson, Fillmore, and Mel'čuk

The formal study of grammatical case begins with the Sanskrit grammarian Pāṇini (c. 4th-5th century BCE), whose *Aṣṭādhyāyī* formalized the *Kāraka* theory — a semantic system of deep roles (e.g., agent, patient) linking nouns to verbs in Sanskrit grammar. Etymologically meaning "that which brings about" an action, the Kāraka system expresses world knowledge through rigorous mappings from deep semantic relations to phonological representations via case endings. As later structuralists noted, "Semantics and phonetics have been the basis of all linguistic analysis in the Sanskritic tradition… providing steps for semantic interpretation through carefully crafted morpho-syntactic procedures" [-@jha2021sanskrit; -@kak1987paninian].

This profound insight—that a finite set of relational roles underlies the infinite variety of sentences—was resurrected in modern linguistics by Charles Fillmore's seminal "The Case for Case" (1968), which explicitly acknowledged its Pāṇinian inheritance

The Cross-Linguistic Typological Landscape

Contemporary typological work reveals that the world's languages realize case systems according to a small number of *alignment types*—systematic patterns governing how the core arguments of transitive and intransitive clauses are grouped [@polinsky2015case; @blake2001grammatical; @haspelmath2009universality].

Core Argument Roles

The cross-linguistic comparison rests on three primitives:

Symbol	Role	Definition
S	Sole argument of intransitive	"The child **sleeps**"
A	Agent-like argument of transitive	"**The child** broke the vase"
P	Patient-like argument of transitive	"The child broke **the vase**"

Alignment Systems

Categorical Formalization of Case Systems

Case Categories as Directed Graphs

We define a **case category** $\mathcal{C}$ as a small category where:

- **Objects** are case roles (NOM, ACC, GEN, DAT, INS, LOC, ABL, VOC)
- **Morphisms** are grammatical relations between roles (e.g., "transitive action": NOM $\rightarrow$ ACC)
- **Identity morphisms** represent the reflexive relation of each case role to itself
- **Composition** models the transitivity of grammatical dependencies

This formalization is implemented in our `CaseCategory` class, which uses a directed graph (via NetworkX) as the underlying representation. Each object carries its role enum and optional morphosyntactic features; each morphism carries a relation label and an enriched weight $w \in [0, 1]$. 1 shows the full eight-case standard category.

Case Category: CEREBRUM Standard